

Building a Pose Map

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2. Building a Grid Map

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1. Use `slam_toolbox` to build a grid map and a pose map

[!NOTE]

- A grid map must be available before annotating the `.geojson` road network file
- The pose map is used to enable quick relocalization when annotating waypoints and during navigation, improving the accuracy of global localization and increasing the accuracy of waypoint annotation.

2. Building a Grid Map

- Enter the following command in the terminal

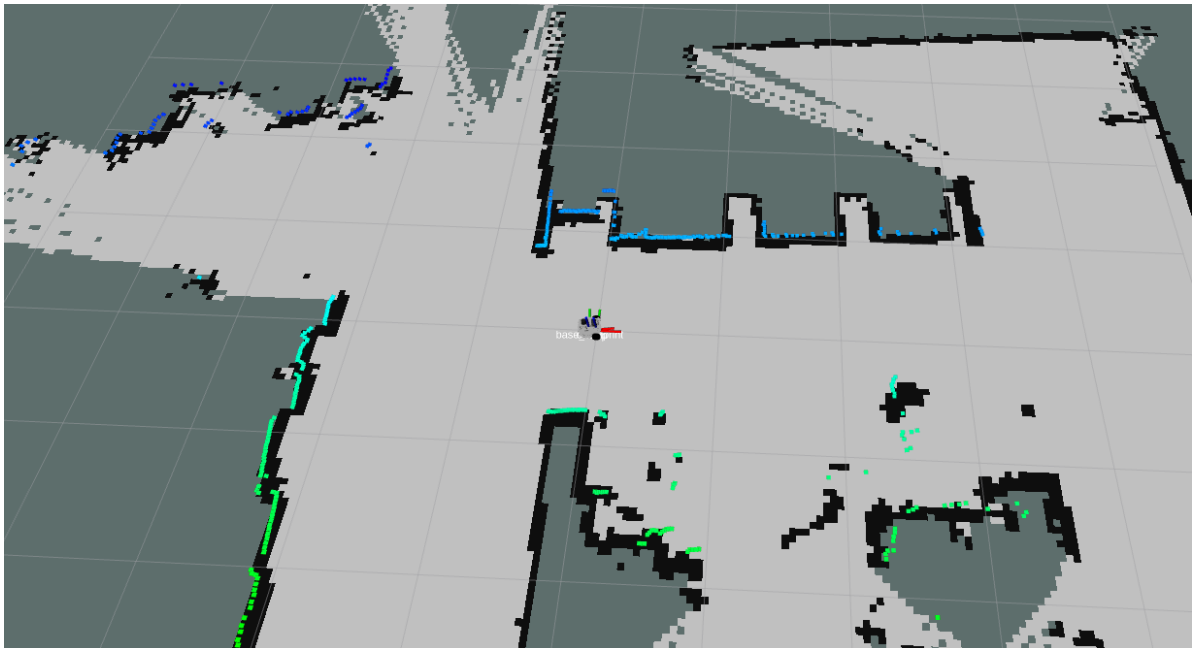
```
ros2 launch microros_v2_bringup slam_toolbox.launch.py
```

- Then start the keyboard teleop node (a gamepad can also be used)

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

```
-----  
Moving around:  
  u    i    o  
  j    k    l  
  m    ,    .  
  
For Holonomic mode (strafing), hold down the shift key:  
-----  
  U    I    O  
  J    K    L  
  M    <    >  
  
t : up (+z)  
b : down (-z)
```

- Move the robot around to build the map



- Save the grid map

```
ros2 launch microros_v2_bringup save_map.launch.py
```

[!TIP]

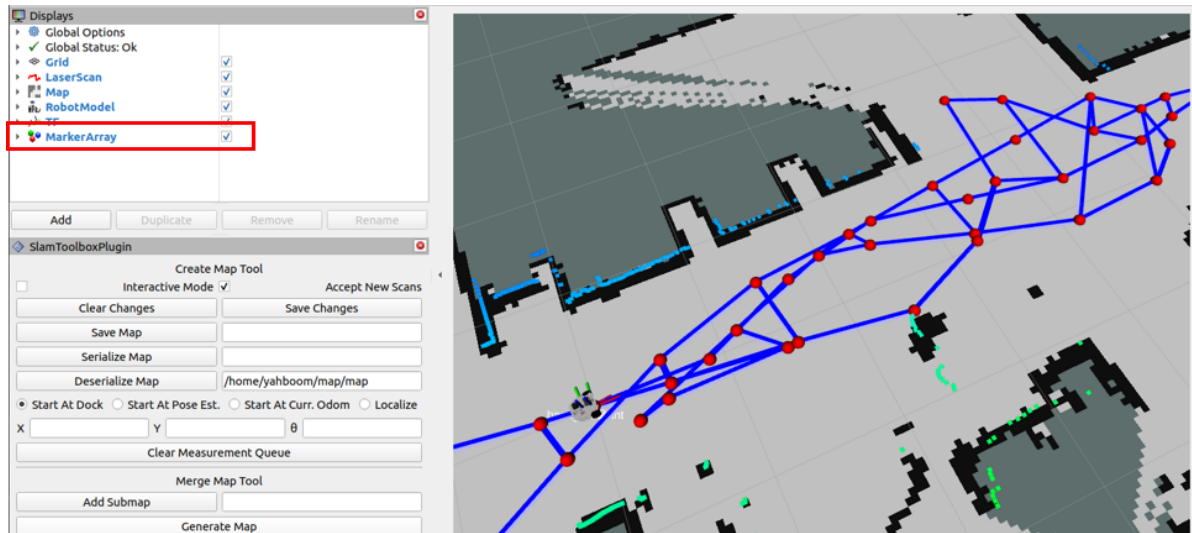
- `map_name` specifies the **map name** of the grid map to be saved. If the launch parameter is not provided, the default map name is `map`. All map files are saved in the `$HOME/map` directory.

The terminal message `Map saved successfully` confirms that the map was saved successfully

```
jetson@yahboom: ~ 82x22
[map_saver_cli-1] map_saver lifecycle node launched.
[map_saver_cli-1] Waiting on external lifecycle transitions to activate
[map_saver_cli-1] See https://design.ros2.org/articles/node_lifecycle.html f
or more information.
[map_saver_cli-1] [INFO] [1763532975.329685271] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1763532975.330069728] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1763532975.341867665] [map_saver]: Saving map from 'map'
topic to '/home/jetson/map/map' file
[map_saver_cli-1] [WARN] [1763532975.359932772] [map_io]: Image format unspecified
. Setting it to: pgm
[map_saver_cli-1] [INFO] [1763532975.361316136] [map_io]: Received a 224 X 598 map
@ 0.05 m/pix
[map_saver_cli-1] [INFO] [1763532975.439369065] [map_io]: Writing map occupancy da
ta to /home/jetson/map/map.pgm
[map_saver_cli-1] [INFO] [1763532975.443013992] [map_io]: Writing map metadata to
/home/jetson/map/map.yaml
[map_saver_cli-1] [INFO] [1763532975.444847063] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1763532975.446514786] [map_saver]: Map saved successfull
y
[map_saver_cli-1] [INFO] [1763532975.449017438] [map_saver]: Destroying
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 50218]
jetson@yahboom:~$
```

- Check the number of pose points (important)
 - Click the `MarkerArray` option in the Display panel in Rviz to view the pose points of the pose map. Make sure there are enough pose points in the map area. If an area has too

few pose points or none at all, the localization performance in that area will be poor during navigation.



- Save the pose map

[!TIP]

- `map_type` specifies the type of map to save. The default is `gridmap`, which saves a grid map; selecting `posegraph` saves the `slam_toolbox` pose map.

The terminal message `slam_toolbox.srv.SerializePoseGraph_Response(result=0)` confirms that the pose map was saved successfully

```
jetson@yahboom: ~ 82x22
jetson@yahboom:~$ ros2 launch m3_bringup save_map.launch.py map_name:=/home/jetson
/map/map map_type:=posegraph
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-11-19
-14-19-41-297598-yahboom-66778
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros2-1]: process started with pid [67023]
[ros2-1] requester: making request: slam_toolbox.srv.SerializePoseGraph_Request(fi
lename='/home/jetson/map/map')
[ros2-1]
[ros2-1] response:
[ros2-1] slam_toolbox.srv.SerializePoseGraph_Response(result=0)
[ros2-1]
[INFO] [ros2-1]: process has finished cleanly [pid 67023]
jetson@yahboom:~$
```

- The map files are saved by default in the `$HOME/map` folder (map.pgm, map.yaml)

